INTEGRATION & DIAGNOSTIC REPORT: T800

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This document outlines the current state of the T800 robot integration project, providing a detailed, step-by-step walkthrough of the simulation environment configuration. Furthermore, it formally reports critical behavioral anomalies encountered while running and training the model within the MuJoCo physics engine.

1. SYSTEM SPECIFICATIONS & ENVIRONMENT ARCHITECTURE

Operating System	`Linux Mint (File System: ext4)`
Network Architecture	`IP Range: 192.168.0.0/24` `Robot IP: 192.168.0.162`
Docker Environment	`Network Mode: host` `Dynamic setup via: t800_docker_setup.tar.gz`
AI Stack (Python)	`Gymnasium (mujoco envs)` `Stable-Baselines3 (PPO - MlpPolicy)`

EngineAI Workspaces	`Native SDK: ~/engineai_robotics_native_sdk/`
	`ROS2 Workspace: ~/source/engineai_workspace/`
	`Asset Context: assets/resource/t800.xml`

2. COMPREHENSIVE SETUP & EXECUTION GUIDE

The following workflow has been consolidated to initialize the environment, compile the MuJoCo engine, configure Reinforcement Learning (RL) dependencies, and establish telemetry streaming with PlotJuggler:

Step 1: Docker Environment Initialization

To prevent conflicts with the host operating system, the entire setup is isolated within the official container. Navigating to the SDK's root directory, the environment is generated and accessed via the provided shortcut, injecting base images and variables dynamically at runtime.

Step 2: Native SDK and ROS2 Compilation

Building the core system is mandatory to generate the ROS2 message (`msg`) interfaces. These interfaces are critical for parsing telemetry data later in PlotJuggler. The root build script is executed strictly within the container.

Step 3: MuJoCo Compilation and Installation

The simulator engine is built using EngineAI's native scripts. Additionally, the required Python libraries are installed to bridge the RL framework (Gymnasium/Stable-Baselines3) with the simulator, enabling policy training.

Step 4: Execution Configuration (Simulation Mode)

Before launching, the control system must be explicitly notified that it is operating in a virtual environment rather than on physical hardware. This is achieved by ensuring the parameter in the first line of `assets/config/t800/mode.yaml` is correctly set to simulation.

Step 5: Integrated Execution (Terminal 1 - MuJoCo)

With ROS2 network environment variables strictly configured, the simulator is executed to initialize the physical simulation of the robot and begin exporting vital telemetry data.

Step 6: Data Visualization (Terminal 2 - PlotJuggler)

A secondary terminal is opened with the exact same environment parameters to launch PlotJuggler. The *"ROS2 Topic Subscriber"* is activated, and the default EngineAI layout (`scripts/plotjuggler/common_data_display.xml`) is loaded to monitor real-time torque and IMU data.

Step 7: State Machine Control (Terminal 3 - Virtual Gamepad)

The robot initializes in a safe `idle` state. A third terminal runs the virtual gamepad control. By activating the interface (F5) and executing keyboard commands (e.g., Q + J, mapping to LB + A), the robot transitions into the `pd_stand` state, generating graphical logs across the motors.

3. MUJOCO SIMULATION ISSUE REPORT

Despite the environment being perfectly configured as per the guide above, critical behavioral flaws have been identified in the T800 model within MuJoCo while developing the Reinforcement Learning policy:

- **Stand-up & Stability Failure:** The simulated T800 model exhibits severe structural difficulties in standing up and maintaining basic upright stability. In most trials, the simulation triggers an early termination due to the robot's center of mass dropping below acceptable thresholds.
- **Missing Control Commands:** The control interface provided for the simulation lacks specific joint commands and balancing parameters. These missing actuations are essential for the controller to effectively stabilize the torso and legs, creating a discrepancy between the simulation and physical hardware expectations.

Request to EngineAI Engineering:
Please review the attached training logs, videos, and Docker configuration (`t800_docker_setup.tar.gz`). I am seeking guidance on whether there is an updated MuJoCo model, a software patch to map the missing commands, or specific PD gains and actuator limits within `t800.xml` that must be adjusted to resolve these stabilization failures.